MedQueue AI — Comprehensive Project & System Architecture

Real-Time ER Clinical Intake, AI Triage & Atomic Priority Queueing Engine

1. Executive Summary & Project Overview

MedQueue AI is a full-stack emergency healthcare queue management system built to eliminate static paper intake forms and waiting room overcrowding. It combines AI voice symptom intake (English & Hindi), automated Emergency Severity Index (ESI Tiers 1–5) classification using Google Gemini API and OpenAI GPT-4o, atomic priority queue re-ordering, and multi-workstation WebSocket synchronization across hospital departments.

2. Full Technology Stack & Frameworks Used

Layer / Category	Technology / Library	Role & Purpose in Project
Frontend UI	React 18 & Vite	Fast component rendering, reactive state, and quick HMR build tool.
Styling & Theme	Tailwind CSS & Glassmorphism	Warm daylight clinical aesthetic, responsive grids, vanilla tones (#FFFBF5).
Animations	Framer Motion & Canvas Confetti	Smooth queue transition animations and celebratory discharge feedback.
Icons	Lucide React	Medical visual indicators (Stethoscope, ShieldAlert, Mic, Heart, Vitals).
Backend Server	Node.js & Express.js	REST API routing, request handling, CORS configuration, environment loading.
Real-Time WebSockets	Socket.io	Instant bi-directional event streaming across Nurse, Doctor & Lobby boards.
AI Triage Models	Google Gemini 2.5 Flash & OpenAI GPT-4o-mini	Natural language symptom analysis, ESI 1-5 classification, red-flag detection.
Schema Validation	Zod	Strict TypeScript/JS runtime schema validation for structured AI outputs.
Database & Storage	MongoDB Mongoose & In-Memory Store	ACID transaction queue re-ordering + zero-setup local memory store.
Voice Processing	Web Speech API & HTML5 MediaRecorder	Continuous microphone speech recognition with real-time text stream.

3. System Workflow & Data Flow Architecture

Step-by-Step Data Flow:

1. **Intake Phase:** Patient speaks or taps symptoms at the Voice Intake Kiosk.
2. **AI Evaluation Phase:** Backend passes input transcript to Google Gemini / OpenAI AI or Local Rule Fallback Engine.
3. **Atomic Queue Re-Ordering:** Patient is assigned ticket ID & rank. ESI 1 & 2 emergency patients automatically jump to Rank #1.

4. **WebSocket Broadcast:** Socket.io pushes updated queue state to Nurse Station, Doctor Workbench, and Lobby Board instantly.

5. **Clinical Action Phase:** Nurse records vitals & assigns exam room; Doctor completes consultation & discharges patient.

4. Complete Module-by-Module Functionality Guide

A. Voice & Symptom Intake Kiosk (/intake)

• **Dual-Language Support:** Switch between English (US) and Hindi (Hindi) for voice intake.

• **Continuous Voice Speech:** Microphone button transcribes spoken words live into the clinical description box.

• **Audio Waveform:** Live animated canvas providing visual mic feedback while recording.

• **Tap-to-Add Disease Chips:** 1-click symptom chips (Chest Pain, Shortness of Breath, High Fever, Migraine, Abdominal Pain, Bleeding, Trauma, Cough).

• **Custom Disease Entry:** Type specific conditions (Dengue, Asthma) and press '+ Add to Box'.

• **AI ESI Classification:** Generates ESI 1-5 rating, chief complaint, red flags, protocol action, and doctor handling time.

B. Nurse Command & Triage Station (/nurse)

• **Live Atomic Queue List:** Displays patients ordered by rank. ESI 1 & 2 emergency patients appear at top with pink/orange glowing badges.

• **Vital Signs Telemetry:** Record Heart Rate (bpm), Blood Pressure (mmHg), SpO2 (%), and Temperature (°F).

• **Manual ESI Override:** Nurses can adjust ESI level tier (1-5) based on clinical observations.

• **Exam Room Assignment:** Assign trauma bays or exam rooms and transition patient to consultation status.

C. Doctor Workbench (/doctor)

• **Active Consultation View:** Physician screen displaying patient vitals, AI summary, red flags, and recommended protocol.

• **Physician Clinical Notes:** Text area for documenting diagnoses, treatment plans, and prescriptions.

• **1-Click Discharge:** Completes consultation, displays celebratory feedback, and automatically calls next priority candidate.

D. Public ER Lobby Display Board (/lobby)

• **Now Serving Cards:** Large display showing ticket numbers, patient names, assigned room numbers, and attending doctors.

• **Emergency Alert Banner:** Animated top banner warning waiting area occupants whenever an ESI 1 or ESI 2 patient enters triage.

E. Telemetry & Analytics Dashboard

• Real-time processing statistics: Total patients intake volume, average handling time, ESI level distribution bar chart, and red flag metrics.

5. Zero-Credit & Fallback Resilience Engine

MedQueue AI features an automatic **Local Clinical Rule Engine**. If your AI API key runs out of credits or quota (HTTP 429), the backend catches the error gracefully and runs local keyword scanning to calculate ESI levels, red flags, and queue rank. **100% of website functions continue operating seamlessly with zero downtime!**

6. Project Links & Repository

• **GitHub Repository:** https://github.com/Samar-111/MedQueue-AI

• **Backend Deployment:** Render / Railway (Root Directory: server)

• **Frontend Deployment:** Vercel / Netlify (Root Directory: client)